

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Solution: Pre-Tutorial Assignment - 9

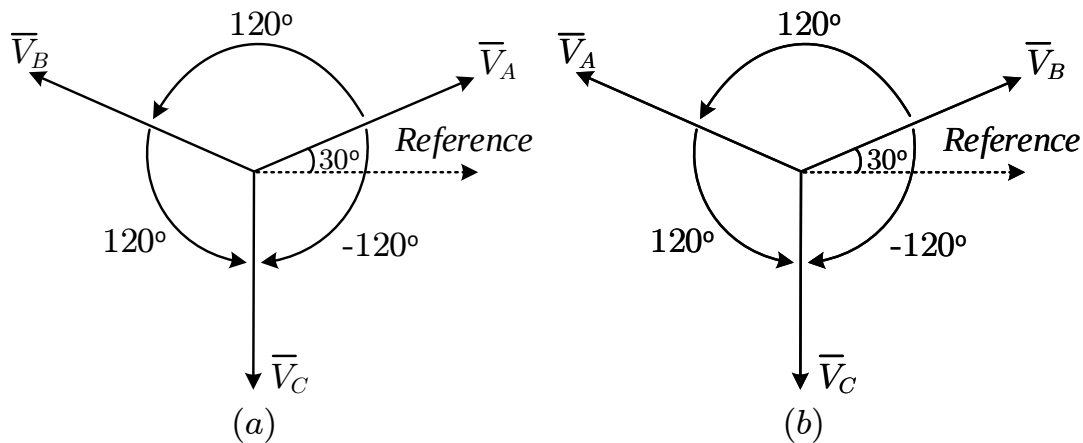
Date: Oct 19, 2023

Marks: 2

1. Given that $\bar{V}_B = 110 \angle 30^\circ \text{ V}$.

(A) Find $\bar{V}_A$ and $\bar{V}_C$, assuming a positive (*abc*) sequence.

(B) Which is the correct phasor representation?



Solution:

Part A:

As, a positive (*abc*) sequence is assumed here,

$$\bar{V}_A = 110 \angle (30 + 120)^\circ \text{ V} = 110 \angle 150^\circ \text{ V}$$

$$\bar{V}_C = 110 \angle (30 - 120)^\circ \text{ V} = 110 \angle -90^\circ \text{ V}$$

Part B: (b)

That's it.